

BIOLOGY

8875/01

Paper 1 Multiple Choice

12 September 2008

1 hour

Additional Materials: Multiple Choice Answer Sheet

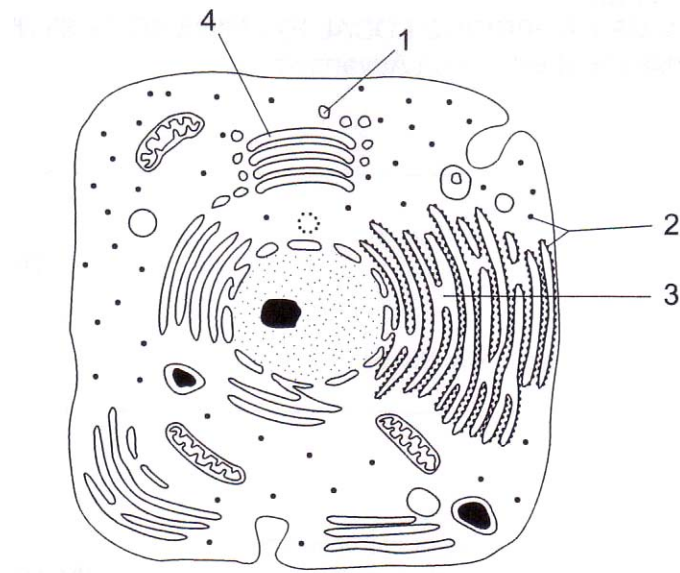
READ THESE INSTRUCTIONS FIRST

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

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1. The diagram shows the ultrastructure of an animal cell.



What are the functions of the labelled structures?

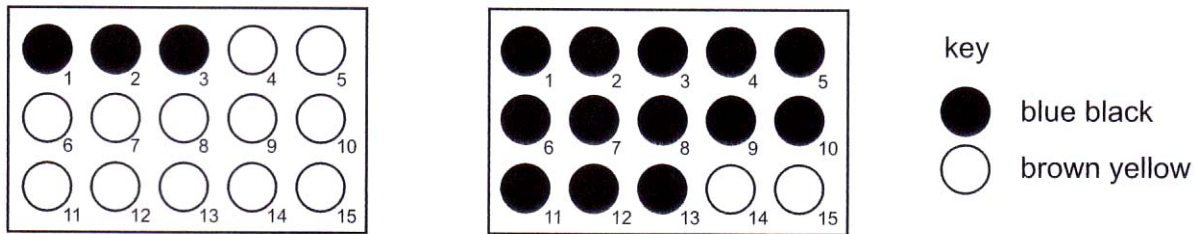
	Synthesizing proteins from amino acids	Transporting proteins	Processing proteins	Secreting digestive enzymes
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

2. From which cell organelle are nucleic acids absent?

- A** chloroplast
- B** Golgi apparatus
- C** mitochondrion
- D** ribosome

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3. An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	temperature / °C	time / s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

4. In active mammalian muscle, the oxidation of 1 g of fat requires 2 litres of oxygen and approximately 2 cm³ of water is produced. For each litre of oxygen used, 1 cm³ of the water produced is lost in the expired air. What is the net water gain to the body from the complete oxidation of 100 g of fat?

- A** 0
- B** 10 cm³
- C** 100 cm³
- D** 200 cm³

5. Which statement about triglycerides is correct?

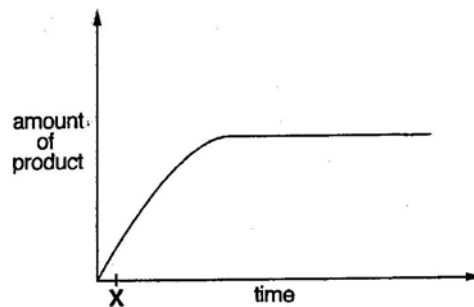
- A They are made up of three fatty acids combined with glycogen.
- B They are more saturated with hydrogen compared with phospholipids.
- C They form a bilayer in the membranes of cells.
- D They have a lower ratio of oxygen to carbon compared with carbohydrates.

6. Which sequence shows the correct order of increasing energy content of molecules?

least $\longrightarrow$ most

- | | | | |
|---|---------|---------|---------|
| A | ATP | starch | sucrose |
| B | ATP | sucrose | starch |
| C | sucrose | starch | ATP |
| D | starch | sucrose | ATP |

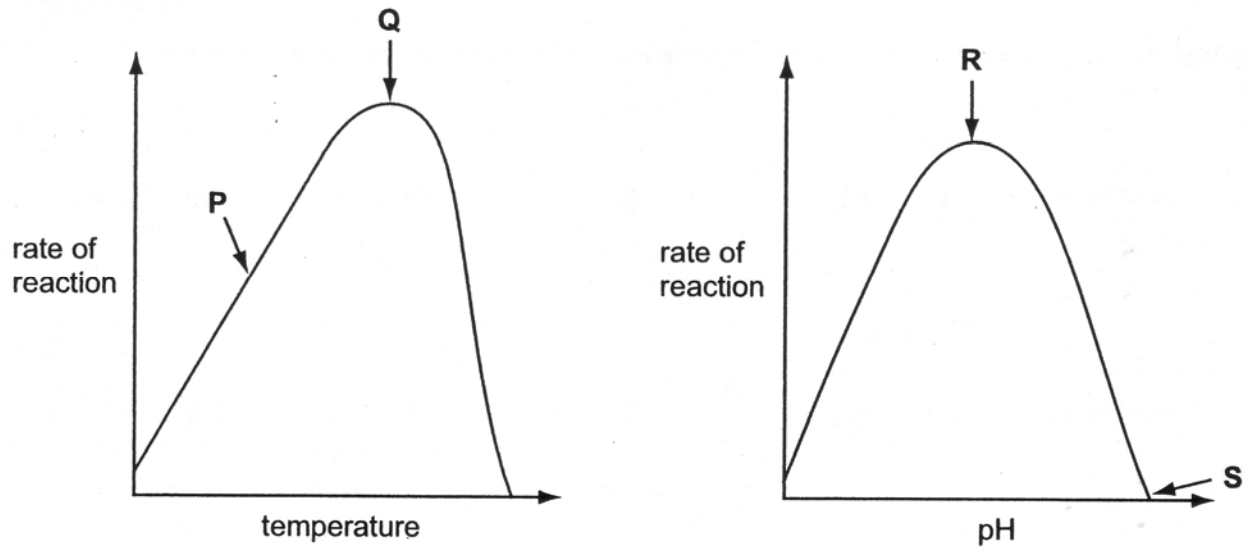
7. The graph shows the course of an enzyme-catalysed reaction at 30° C.



What is true at time X?

- A Most enzyme molecules will have free active sites.
- B The number of available substrate molecules is high.
- C The number of enzyme-substrate complexes is low.
- D The rate remains the same if more enzyme is added.

8. The graphs show the effects of temperature and pH on enzyme activity.



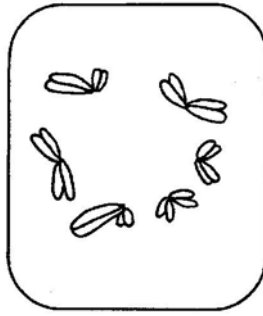
Which statement explains the enzyme activity at the point shown?

- A At P, hydrogen bonds are formed between enzyme and substrate.
- B At Q, the kinetic energy of enzyme and substrate is highest.
- C At R, peptide bonds in the enzyme begin to break.
- D At S, the substrate is completely denatured.

9. Which combination of chromosomes would result in a zygote showing polyploidy, if the parental chromosome number is 36?

	Sperm	Ovum
A	9	9
B	18	18
C	18	19
D	18	36

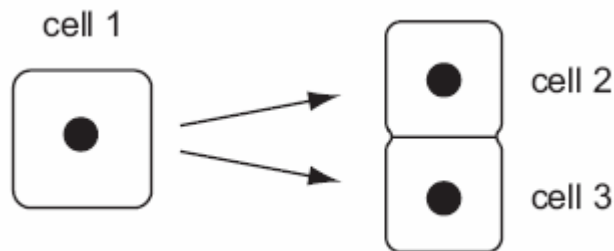
10. What does this represent?



- A A cell during metaphase.
- B A stage of mitosis in a polyploidy cell.
- C Homologous chromosomes preparing to cross over.
- D Pairing of homologous chromosomes.

11. Human cells contain 46 chromosomes.

The diagram shows a human cell in prophase of mitosis (cell 1) and the daughter cells just after telophase (cells 2 and 3).



How many DNA molecules are there in the nucleus of cell 1 and cell 2?

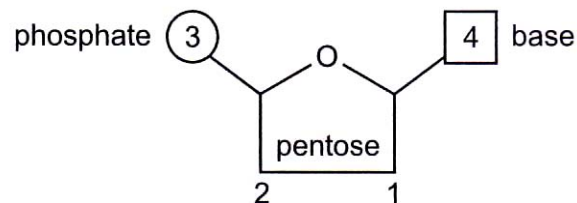
	cell 1	cell 2
A	46	23
B	46	46
C	92	23
D	92	46

12. Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria (generation 1) were then grown in a medium containing only normal nitrogen (^{14}N) for two more generations (generations 2 and 3). The percentage of cells containing ^{15}N in each generation was estimated.

What will be the percentage of cells containing ^{15}N in generations 2 and 3?

	Percentage of cells containing ^{15}N	
	generation 2	generation 3
A	100	100
B	100	50
C	50	50
D	50	25

13. The diagram shows the structure of a nucleotide.



One type of polynucleotide consists of a single strand, twisted into a clover leaf shape. Which is the type of pentose of this molecule and the bonding between the nucleotides?

	Pentose	Bonding between nucleotides
A	deoxyribose	Between positions 1 and 3
B	deoxyribose	Between positions 2 and 3
C	ribose	Between positions 4 and 4
D	ribose	Between positions 2 and 3

14. One individual, organism **P**, in a sexually-reproducing population of mammals has two forms of a protein in its heart muscle. The normal form is identical to that in most other members of the population, while the variant form differs by one amino acid. The diagram shows the amino acid sequences in part of the normal protein and in the same part of the variant protein.

Normal protein	Variant protein
-gly-lys-lys-gly-glu-	-glu-lys-gly-gly-glu-

What could account for the presence of the normal protein and the variant protein in P?

- 1 The sequence of amino acids was altered in a cell in the heart muscle of P's mother.
- 2 The sequence of amino acids was altered in a cell in the ovaries of P's mother.
- 3 The sequence of bases was altered in the DNA in a cell in the heart muscle of P.
- 4 The sequence of bases was altered in the DNA in a cell in an ovary of P.
- 5 The sequence of bases was altered in the DNA in a cell in a testis of P's grandfather.

- A** 1, 2 and 4 only
- B** 1 and 3 only
- C** 2, 4 and 5 only
- D** 3 and 5 only

15. A peptide consists of ten amino acids of four different kinds. What is the theoretical minimum number of tRNA molecules required to translate the mRNA for this peptide?

- A** 4
- B** 10
- C** 12
- D** 30

16. In most organisms, all possible triplets of bases in DNA code for an amino acid, except for the coding strand 'stop' triples ATT, ATC and ACT. The coding strand is complementary to mRNA.

In the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid, pyrrolysine.

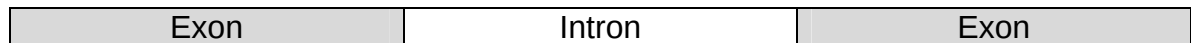
The **non-coding** strand of a section of DNA in *M. barkeri* has the following sequence of bases.

TTTTTATTGTATTACTAGTGTTAATGA

How many amino acids can be made into a peptide by *M. barkeri* from the **coding** strand of this region of DNA?

- A** 5 **B** 6 **C** 7 **D** 9

17. The diagram shows a gene in a human cell.



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A** Activation of the gene by a steroid hormone
- B** Attachment of mRNA to a ribosome
- C** Binding of RNA polymerase to DNA
- D** Splicing of pre-mRNA

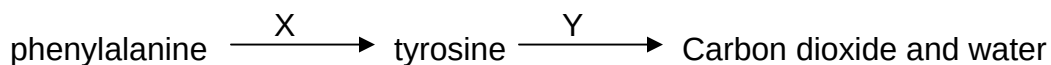
18. Which of the following is a difficulty in getting prokaryotic cells to express eukaryotic genes?

- A** The signals that control gene expression are different and prokaryotic promoter regions must be added to the vector.
- B** Prokaryotic cells cannot transcribe introns because their genes do not have them.
- C** The ribosomes of prokaryotes are not large enough to handle long eukaryotic genes.
- D** The RNA splicing enzymes of bacteria are different from those of eukaryotes.

19. Day-old chicks can be difficult to sex, but their down feathers reveal a sex-linked dominant character called 'barred'. Female birds are XY and males are XX. Which cross would ensure correct sexing of the resultant chicks?

	Female	Male
A	barred	homozygous barred
B	barred	homozygous unbarred
C	unbarred	heterozygous barred
D	unbarred	homozygous barred

20. The following reaction sequence occurs in humans.



Phenylketonuria (PKU) is a disease caused by an enzyme deficiency in step X, and alkaptonuria (AKU) is due to an enzyme deficiency in step Y. Both conditions are rare and caused by recessive alleles. A person with PKU marries a person with AKU. Which phenotypes would be expected for their children?

- A** All normal
- B** All AKU only
- C** All PKU only
- D** All both AKU and PKU

21. The table shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1 st parent	Genotype of 2 nd parent	
	$I^A I^A$	$I^A I^O$
$I^A I^O$	A	A and O
$I^B I^O$	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria. What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

- A** 1 in 16
- B** 1 in 8
- C** 3 in 16
- D** 3 in 8

22. Which two products result from the photoactivation of chlorophyll within a chloroplast?

- A** ATP and reduced NADP
- B** ATP and H_2O
- C** CO_2 and ATP
- D** Reduced NADP and H_2O

23. Six tubes were set up as shown in the table.

tube	contents
1	Glucose + homogenized animal cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells
5	Pyruvate + mitochondria
6	Pyruvate + cytoplasm lacking organelles

After incubation, each sample was analysed to determine the presence of carbon dioxide and lactate. In which tubes is lactate most likely to be present?

- A** 1 and 3 only
- B** 2, 3, 5 and 6 only
- C** 3 and 6 only
- D** 4, 5, and 6 only

24. Which substances enter and leave a mitochondrion during aerobic respiration?

	enters	leaves
A	acetyl Co-A	carbon dioxide
B	ADP	lactate
C	glucose	reduced NAD
D	pyruvate	ATP

25. Some organisms change the composition of their membranes to maintain membrane fluidity when temperature changes.

Which changes occur in the composition of membranes following a change to low temperature?

- A** Increase phospholipids, decrease cholesterol
- B** Increase saturated fatty acids, increase cholesterol
- C** Increase saturated fatty acids, decrease cholesterol
- D** Increase unsaturated fatty acids, increase cholesterol

26. Two populations of a given species could only evolve into two distinct species if they were subjected to

- A** disruptive selection.
- B** genetic isolation.
- C** geographical isolation.
- D** stabilising selection.

27. Some statements concerning evolution are listed:

1. Characteristics acquired during an organism's lifetime are passed to its offspring.
2. Individuals in a sexually reproducing population are different.
3. More offspring are produced than can possibly survive to sexual maturity.
4. Offspring tend to resemble their parents.
5. The fossil record shows that many species have become extinct.

Which of these statements form the basis of Darwin's theory of evolution by natural selection?

- A** 1 3 4
- B** 1 4 5
- C** 2 3 4
- D** 2 4 5

28. If the first three nucleotides in a six-nucleotide restriction site are CTG, what would the next three nucleotides most likely be?

- A** AGG
- B** GTC
- C** CTG
- D** CAG

29. At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60°C.

What explains why the denatured DNA strands anneal with primers and **not** each other?

- A** The primers are shorter and anneal more easily.
- B** The primers anneal only to the 3' end of denatured DNA.
- C** The primer concentration exceeds the denatured DNA concentration.
- D** The temperature prevents the denatured DNA from annealing together.

30. Which statement describes a unique feature of stem cells in adult humans?

- A** They can differentiate into different cell types.
- B** They can undergo meiosis.
- C** They have a complete human genome.
- D** They have multiple copies of active genes.

Answers for H1 P1 Anderson Junior College Biology Prelims

1	C	7	B	13	D	19	B	25	D
2	B	8	A	14	D	20	A	26	B
3	B	9	D	15	A	21	C	27	C
4	A	10	A	16	C	22	A	28	D
5	D	11	D	17	D	23	C	29	C
6	B	12	B	18	A	24	D	30	A